



THE UNIVERSITY OF
MELBOURNE

Marketing Research

Week 7





Topics for today

1. Recap: CI and sample size
2. Hypothesis Testing
3. Testing for Difference: t-tests
4. Activities on SPSS
5. You should receive the feedback for your questionnaire design by April 22 (T) at 11:59 pm



Recap: Describe data

Info level / adv. methods ↓	Scale Level	Measure of Central Tendency	Measure of Variability	Type
	Nominal	Mode	(Frequency)	non-metric
	Ordinal	Median	Range	
	Interval	Mean	Standard Deviation	Metric
	Ratio			



Recap: Confidence Intervals in MR

How confident are we in what we found?



What Is a Confidence Interval?



A confidence interval (CI) is a range of values we believe contains the true value.



It helps generalize from a sample to the whole population.



CI tells us: "If we repeated this many times, this range would contain the truth most of the time."

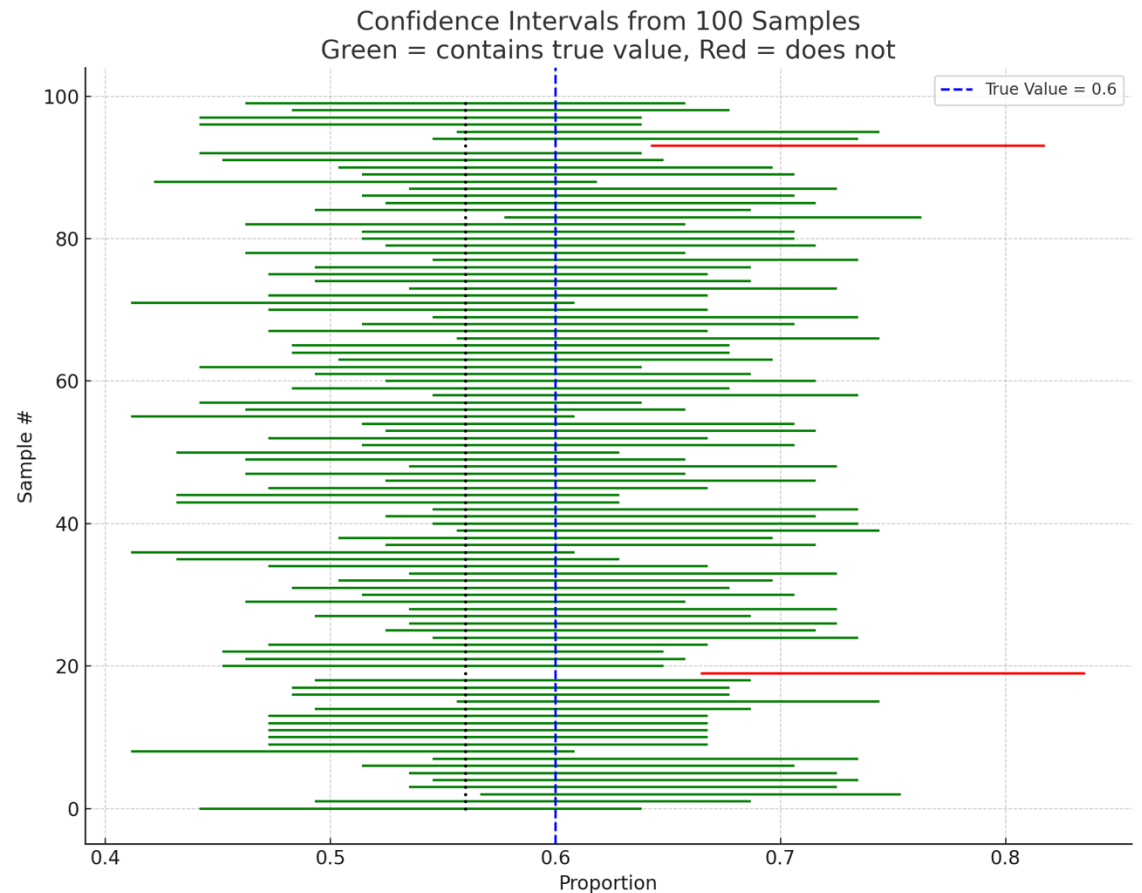


Intuition: Like a target, the CI is the area where most of your arrows (estimates) land.



Example

- **Survey:** 100 Melbournian consumers, 60% prefer Woolworths over Coles
- **CI:** $60\% \pm 5\%$ at 95% confidence $\rightarrow$ [55%, 65%]
- **Meaning:** We are 95% confident the true population proportion is in this range.
- **Misconception:** It's not a 95% chance — it's 95% confidence across many samples.



It shows 100 confidence intervals from simulated samples:

- **Green lines:** intervals that contain the true value (0.6)
- **Red lines:** intervals that *miss* the true value (around 5%, as expected at 95% confidence)



What Affects CI Width?

- Sample size $\uparrow \rightarrow$ CI narrower
- Data variability $\uparrow \rightarrow$ CI wider
- Confidence level $\uparrow \rightarrow$ CI wider

Note:

- Confidence level = how sure we want to be.
- Example: 95% confidence = 95 out of 100 samples will contain the true value.
- Common levels: 90%, 95%, 99%



Why CI matters in Marketing

Ad testing: Did the campaign improve brand recall (on a Y/N scale)?

- 2023 mobile ad campaign for a new ride-hailing app in Vietnam.
- Survey of 400 users: 58% remembered the brand.
- 95% CI = $58\% \pm 4.8\% = [53.2\%, 62.8\%]$.
- Implies majority of market recalls brand (> 50%) → strong campaign reach.
- Source: App Annie & YouGov Vietnam, Mobile Ad Recall Study, 2023

Ad testing: A better brand recall (on a 5-point scale) from the campaign? Is it significantly different from neutral?

- 2022 survey of 250 Australians after digital health campaign.
- Average recall score = 4.2 (on 5-point scale), CI = 4.2 ± 0.15 (95% confidence).
- 95% confident true recall is between 4.05 and 4.35.
- Even lower bound (4.05) > neutral (3.0) → campaign likely effective.
- Source: Australian Department of Health and Aged Care, 2022



Confidence Interval Formula

Used when you're estimating a percentage or proportion (e.g., % of customers who prefer a brand).

$$\text{Formula: CI} = \hat{p} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Where:

- $\hat{p}$ = sample proportion (e.g., 0.58)
- n = sample size
- z = z-score for desired confidence (1.96 for 95%)

Margin of Error = $z \times$ standard error of the proportion

Used when you're estimating an average (e.g., average satisfaction score).

$$\text{Formula: CI} = \bar{x} \pm z \frac{\sigma}{\sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- σ = standard deviation (s for sample SD)
- n = sample size
- z = z-score for desired confidence (1.96 for 95%)

Margin of Error = $z \times$ standard error of the mean



What Is a Z-score in a Confidence Interval?

The z-score reflects how many standard errors we go out from the estimate to build the interval.

It depends on the confidence level you want:

- 90% confidence $\rightarrow z = 1.645$
- 95% confidence $\rightarrow z = 1.96$
- 99% confidence $\rightarrow z = 2.576$

Higher confidence $\rightarrow$ wider interval $\rightarrow$ larger z-score

Z-score assumes a normal distribution (central limit theorem helps this for large samples)



Summary: What Goes into a Confidence Interval?

1. Point estimate: What you got from your sample (mean or proportion).
2. Standard error (SE): How much your estimate is expected to vary from sample to sample.
3. Z-score: Reflects your desired confidence level (e.g., 95%).

$$CI = \text{Point Estimate} \pm z \times SE$$

- Narrow CI → larger sample, lower variability, or lower confidence
- Wide CI → small sample, high variability, or high confidence



From CI to Sample Size – Proportion

Start from the confidence interval formula for a proportion.

$$\text{Margin of Error } (E) = z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Rearrange to solve for sample size n:

$$n = \hat{p}(1 - \hat{p}) \left(\frac{z}{E} \right)^2$$

Choose:

- Confidence level $\rightarrow$ z-score (1.96 for 95%)
- Margin of error (e.g., $\pm 4.8\%$)
- Estimated proportion (e.g., $p = 0.58$ from Vietnam campaign)

This helps us plan how many responses we need to achieve the desired precision.



Sample Size Planning Example

Vietnam Ad Campaign

From the Vietnam ride-hailing ad campaign:

- Estimated proportion $p = 0.58$
- Target MoE = $\pm 4.8\% = 0.048$
- Confidence level = 95% $\rightarrow z = 1.96$

Apply formula: $n = (1.96^2 \times 0.58 \times 0.42) / 0.048^2 \approx 405$

Conclusion: Need at least 405 responses for $\pm 4.8\%$ MoE at 95% confidence

The actual survey had $n = 400 \rightarrow$ close to ideal, results are reliable



From CI to Sample Size – Mean

Start from the CI formula for a population mean.

$$\text{Margin of Error } (E) = z \frac{\sigma}{\sqrt{n}}$$

Rearranged to solve for sample size n :

$$n = \left(\frac{(z)(\sigma)}{E} \right)^2$$

Use case: Australia digital health campaign

- MoE = 0.15, $s = 0.6$ (to proxy for population σ), $z = 1.96 \rightarrow n \approx 62$



From CI to Sample Size – Mean

Rearranged to solve for sample size n:

$$n = (z \times s / \text{MoE})^2$$

Use case: Australia digital health campaign

- MoE = 0.15, s = 0.6, z = 1.96 $\rightarrow$ n $\approx$ 62



Exercise 1: Confidence Interval for Proportion (Consumer Preference)

A market research firm surveyed **500 online shoppers in Malaysia** and found that **62% preferred using mobile apps over websites** for shopping.

Question:

1. Construct a **95% confidence interval** for the proportion of all online shoppers in Malaysia who prefer mobile apps.
2. Interpret the interval in plain language.
3. If the company wants to reduce the margin of error to $\pm 3\%$, what **minimum sample size** would be needed at 95% confidence?



Exercise 2: Confidence Interval for Mean (Satisfaction Score)

A digital health platform in Singapore tested a new interface. After the pilot launch, a **sample of 40 users** rated their satisfaction on a 1–7 scale. The **sample mean was 5.4** and the **standard deviation was 1.2**.

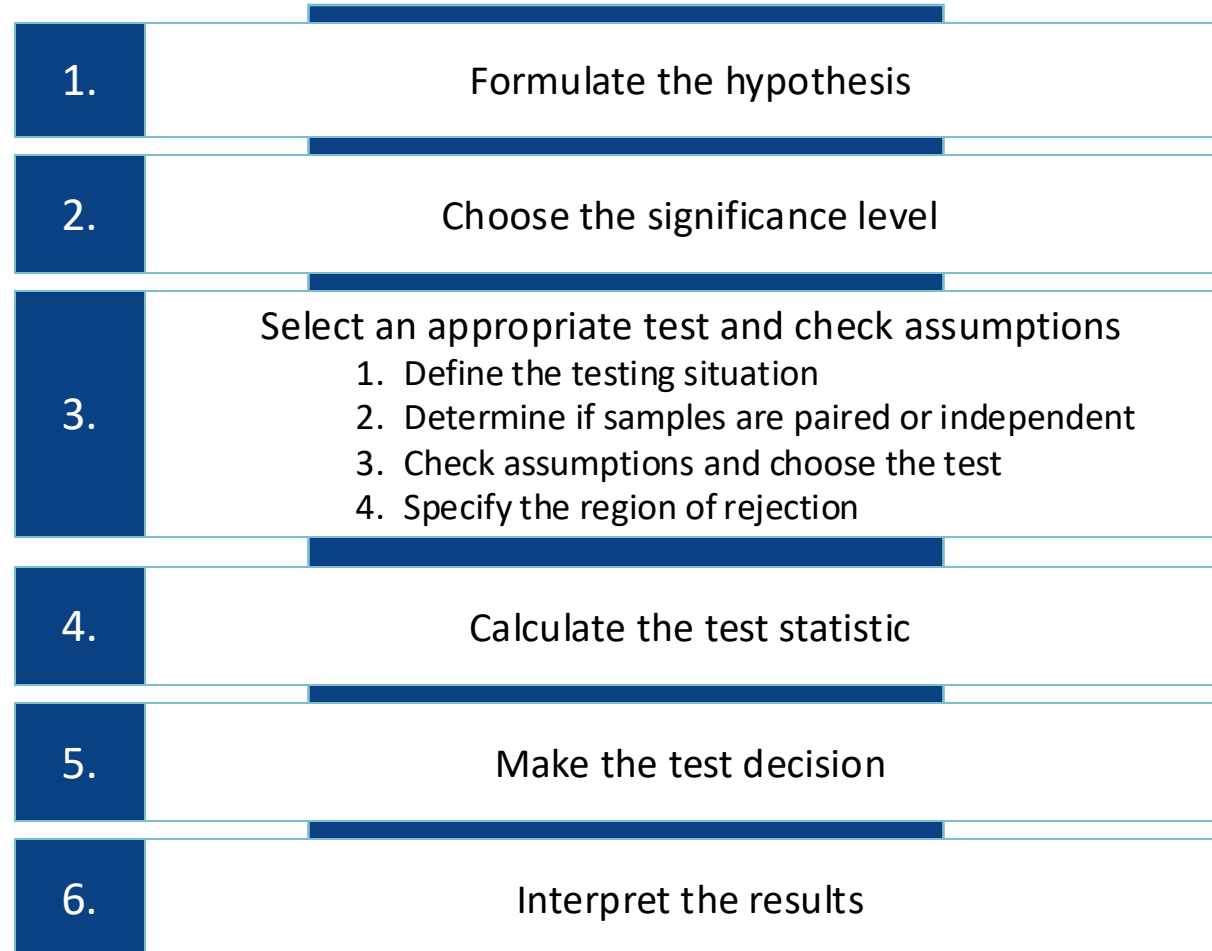
Question:

1. Construct a **95% confidence interval** for the true mean satisfaction score.
2. Interpret the interval in context.
3. The team wants to estimate the average with a **margin of error of ± 0.3** . What **sample size** is required at 95% confidence?



Hypothesis testing

Principles of Hypothesis Testing



1. Formulate the hypothesis

Hypothesis testing starts with the formulation of a null and alternative hypothesis

	Null hypothesis H_0	Alternative hypothesis H_1
Definition	A statement expecting no difference or effect	A statement expecting a difference or effect (opposite of the null hypothesis)
Example	There's no difference in mean sales between stores that installed a point of sale display versus those that installed a free tasting stand (statistically, average sales for point of sale display = average sales of free tasting stand)	There's a difference in mean sales between stores that installed a point of sale display versus those that installed a free tasting stand (statistically, average sales for point of sale display $\neq$ average sales of free tasting stand)
Outcomes	• Reject the null hypothesis $\rightarrow$ support for alternative hypothesis • Do not reject the null hypothesis $\rightarrow$ The conclusion is that no difference is most likely.	

Note that researchers always refer to the alternative hypothesis when referring to the hypothesis to be tested!



1. Formulate the hypothesis

Example of hypothesis

- “What is your evaluation of brand A?”
- Very negative 1 2 3 4 5 6 7 Very positive
- “How likely are you to purchase brand A?”
- Very unlikely 1 2 3 4 5 6 7 Very likely
- H₀: consumer evaluation does not affect purchase intention
- H₁: consumer evaluation affects purchase intention

12. Are you likely to attend such programs again in the next 6 months?

Not at all Likely					Very Likely
1	2	3	4	5	

2. Please indicate your gender

- ☐ Male
- ☐ Female

- H₀: The intention of consumers to reattend the program does not vary by gender.
- H₁: The intention of consumers to reattend the program varies by gender.

2. Choose the significance level (II)

- **No** type of hypothesis test can validate hypotheses with absolute certainty.
- Since relying on sample data to make inferences and not the whole population, we don't know the true state of H_0 -> some probability of a wrong conclusion
- However, we can (partly) control for this risk by setting an acceptable probability (called significance level α) of erroneously rejecting the null hypothesis
- The probability of a **type I error** is inversely related to that of a **type II error**, so that the smaller the risk of one type of error, the higher the risk of the other!

		True state of H_0	
		H_0 true	H_0 false
Test decision	H_0 rejected	Type I error	✓
	H_0 not rejected	✓	Type II error



3. Select an Appropriate Test and Check Assumptions

Tests for Difference Analysis

Type of Tests	Purpose of Test
One Sample T-Test	Compare the mean of a question answered by one group of subjects against a constant on metric data
Paired Samples T-Test	Compare two means of two different questions answered by the same group of subjects on metric data
Independent Samples T-Test	Compare the mean of a question answered by two different groups of subjects on metric data
ANOVA	Compare the means of a question answered by more than two different groups on metric data



3. Select an Appropriate Test and Check Assumptions

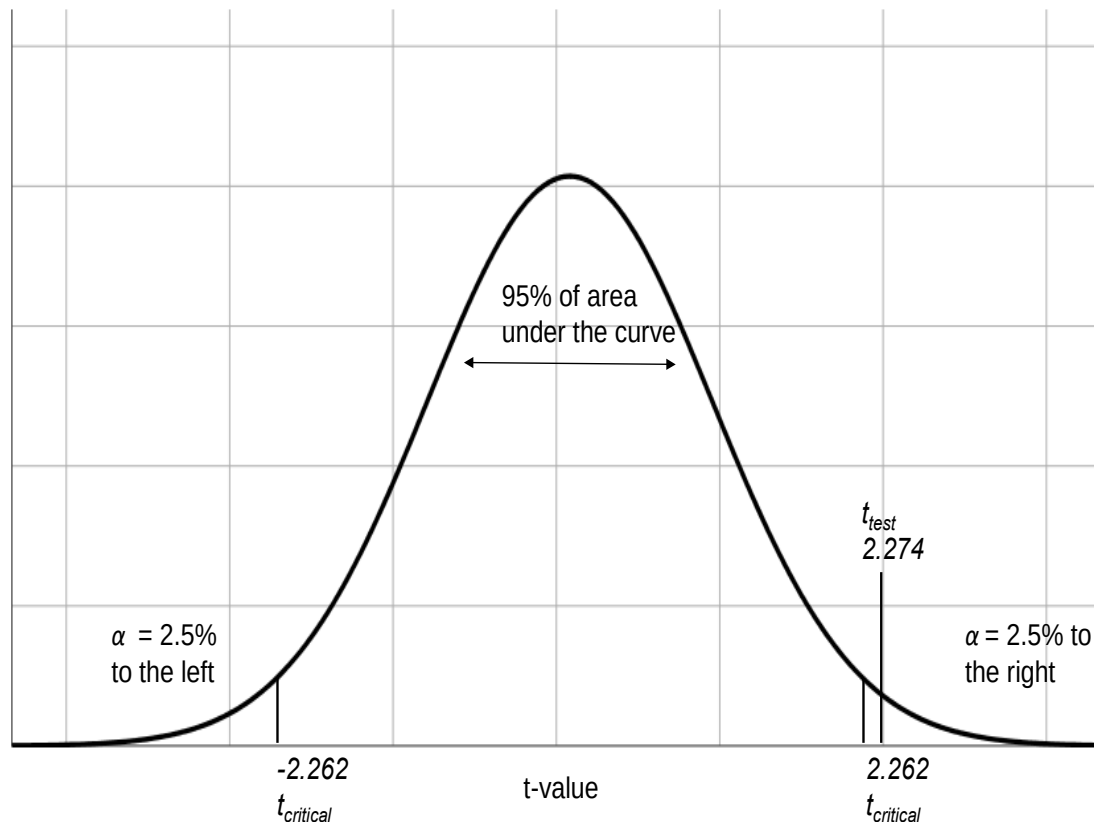
Tests for Associations

Types of Test	Purpose of Test
Chi-Square Analysis (X^2)	Test for relationship between two <u>nominal</u> variables.
Spearman Rank Order Correlation Coefficient	Test for relationship between two variables that are not nominal and that at least one is <u>ordinal</u> .
Pearson Product Moment Correlation Coefficient	Test for relationships between two <u>interval and/or ratio</u> variables.

5. Make the test decision

Relationship between test value, critical value and p-value:

The t-distribution for 9 degrees of freedom
Two-tailed test and $\alpha = .05$



Decision rules:

- SPSS automatically calculates the p-value (= the probability of obtaining a test statistic at least as extreme as the one that is actually observed)
- The p-value can also be interpreted as the probability of erroneously rejecting a true null hypothesis
- The significance level α in contrast is set by the researcher before the test outcome is observed

- ▶ $p\text{-value} \leq \alpha \rightarrow \text{reject } H_0$
- ▶ $p\text{-value} > \alpha \rightarrow \text{do not reject } H_0$



6. Interpret the results

Draw a conclusion, such as “ there is evidence that the point of sale display influenced the number of sales significantly during the week it was installed”

Examples

In this test we hypothesized that consumers are satisfied with the exhibitions in the museum. The result of a one-sample t test shows that the mean of satisfaction towards the museum’s exhibitions ($M = 4.27$) is significantly higher than the test value of 3 ($t_{55} = 6.20$, $p < .001$). Hence our hypothesis was supported.

In this test we hypothesized that visitor’s role affects their tendency to attend art talks. The result of a one-way ANOVA test shows that visitor’s role did affect the intention to attend art talks ($F(3, 246) = 5.413$, $p < .01$). Hence our hypothesis was supported. Further post-hoc analyses show that academic staff and former students are both more likely to attend art talks than current students ($M_{\text{academic}} = 4.13$ vs. $M_{\text{current student}} = 3.46$, $p < .05$; $M_{\text{former student}} = 4.14$ vs. $M_{\text{current student}} = 3.46$, $p < .01$). Also, former students are also more likely to attend art talks than professional staff ($M_{\text{former student}} = 4.14$ vs. $M_{\text{professional staff}} = 3.74$, $p < .05$). Other groups of comparisons are not significant.

In this test we hypothesized that consumer evaluations of different facilities can predict their visit frequency. The results of a Multiple Regression Analysis show that evaluation of opening hours has a significant positive impact on visit frequency ($\beta = 0.18$, $p < .05$), and evaluation of parking ($\beta = -0.18$, $p < .01$) and that of bookshop ($\beta = -0.26$, $p < .01$) have a significant negative impact on visit frequency.



Testing for Difference - One Sample T-Test



One sample t-test

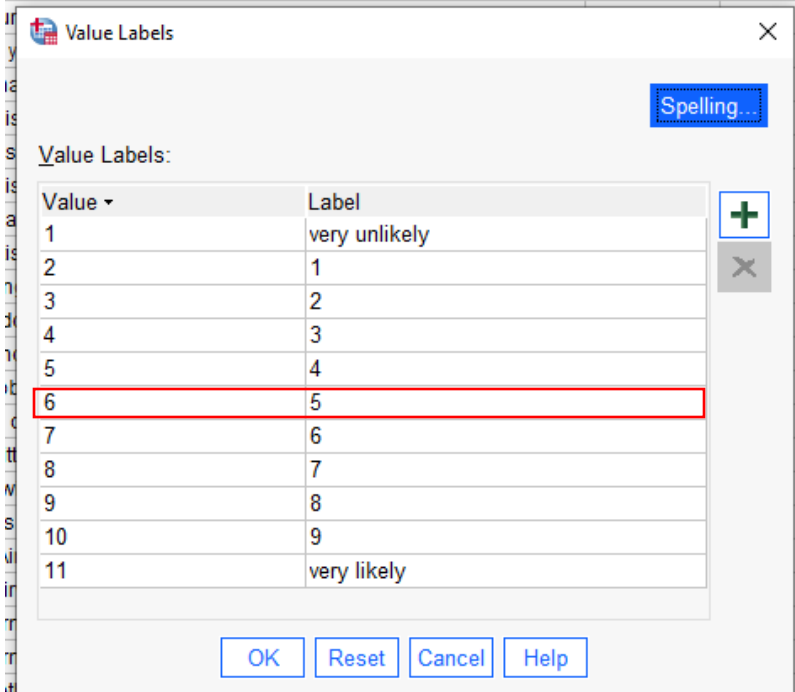
- ❑ Test the mean against a constant test value (the test value is typically the middle value of the scale).

Assumption

- Test variable measured on an interval or ratio scale
- Independence of observation
- $n > 30$

Example

- One group; interval scale -> one sample t test
- Significance level: 5%
- Testing variable: willingness to recommend (nps); Test value: 6
- Hypothesis: The average willingness to recommend the airline is greater than 6.



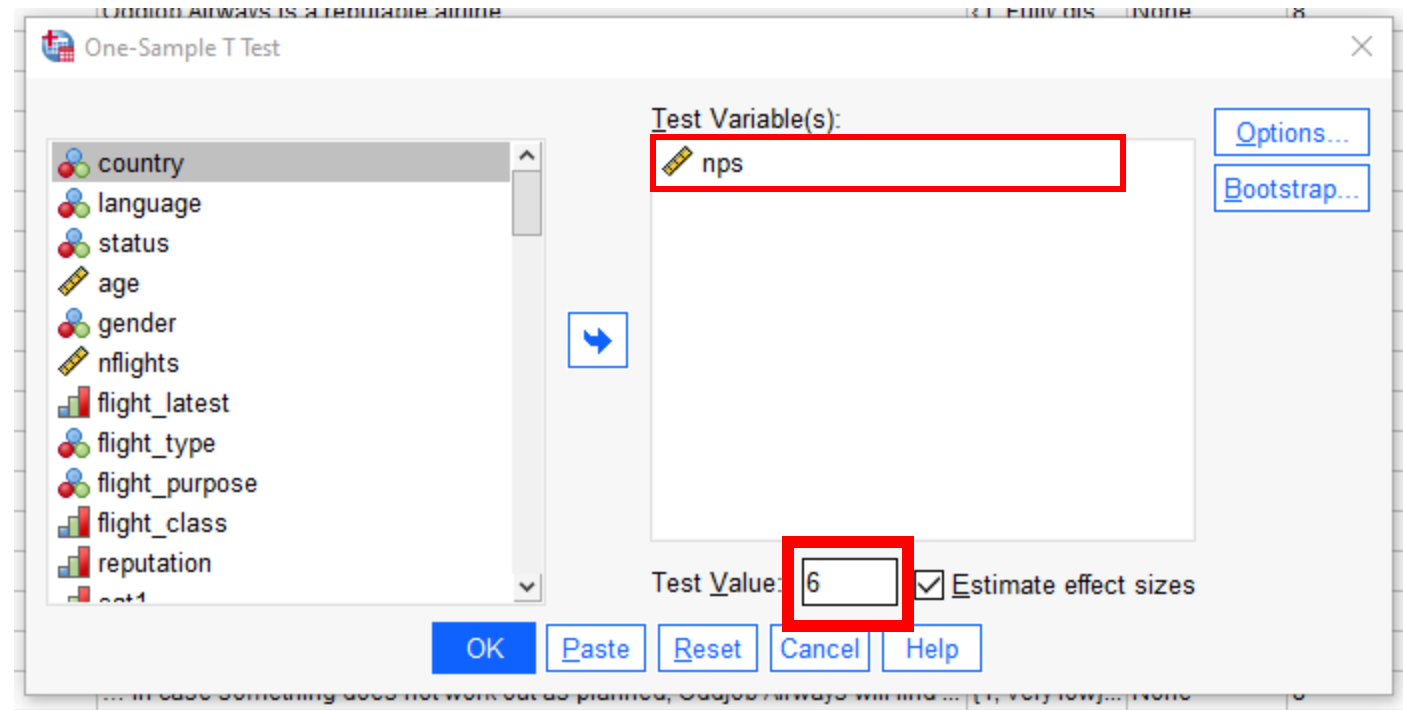
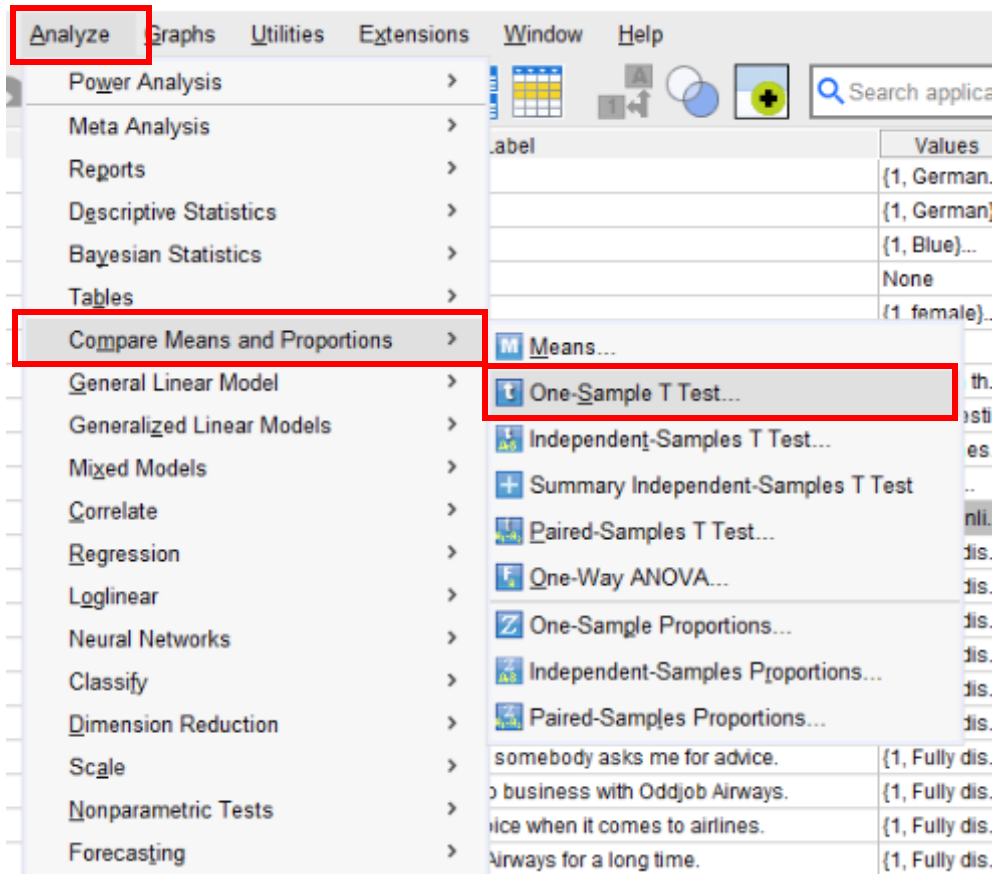
The image shows the 'Value Labels' dialog box in SPSS. It contains a table with 'Value' and 'Label' columns. The values range from 1 to 11. The label for value 1 is 'very unlikely', and for value 11 is 'very likely'. The row for value 6 is highlighted with a red border. There are buttons for 'Spelling...', '+', and '-' on the right side of the table. At the bottom, there are buttons for 'OK', 'Reset', 'Cancel', and 'Help'.

Value	Label
1	very unlikely
2	1
3	2
4	3
5	4
6	5
7	6
8	7
9	8
10	9
11	very likely



One sample t-test

Analyze -> Compare Means and Proportions -> One-sample T Test -> Select the variable of interest -> Enter test value



One sample t-test

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
How likely is it that you would recommend our company to a friend or colleague?	1065	8.28	2.516	.077

One-Sample Test

	t	df	Significance		Mean Difference	95% Confidence Interval of the Difference	
			One-Sided p	Two-Sided p		Lower	Upper
How likely is it that you would recommend our company to a friend or colleague?	29.611	1064	<.001	<.001	2.283	2.13	2.43

- Mean = 8.28 (> test value 6)
- Sig. value (p value) < 0.001 < 0.5
- H1 is supported



One sample t-test

How to report the results of one sample t-test?

Example

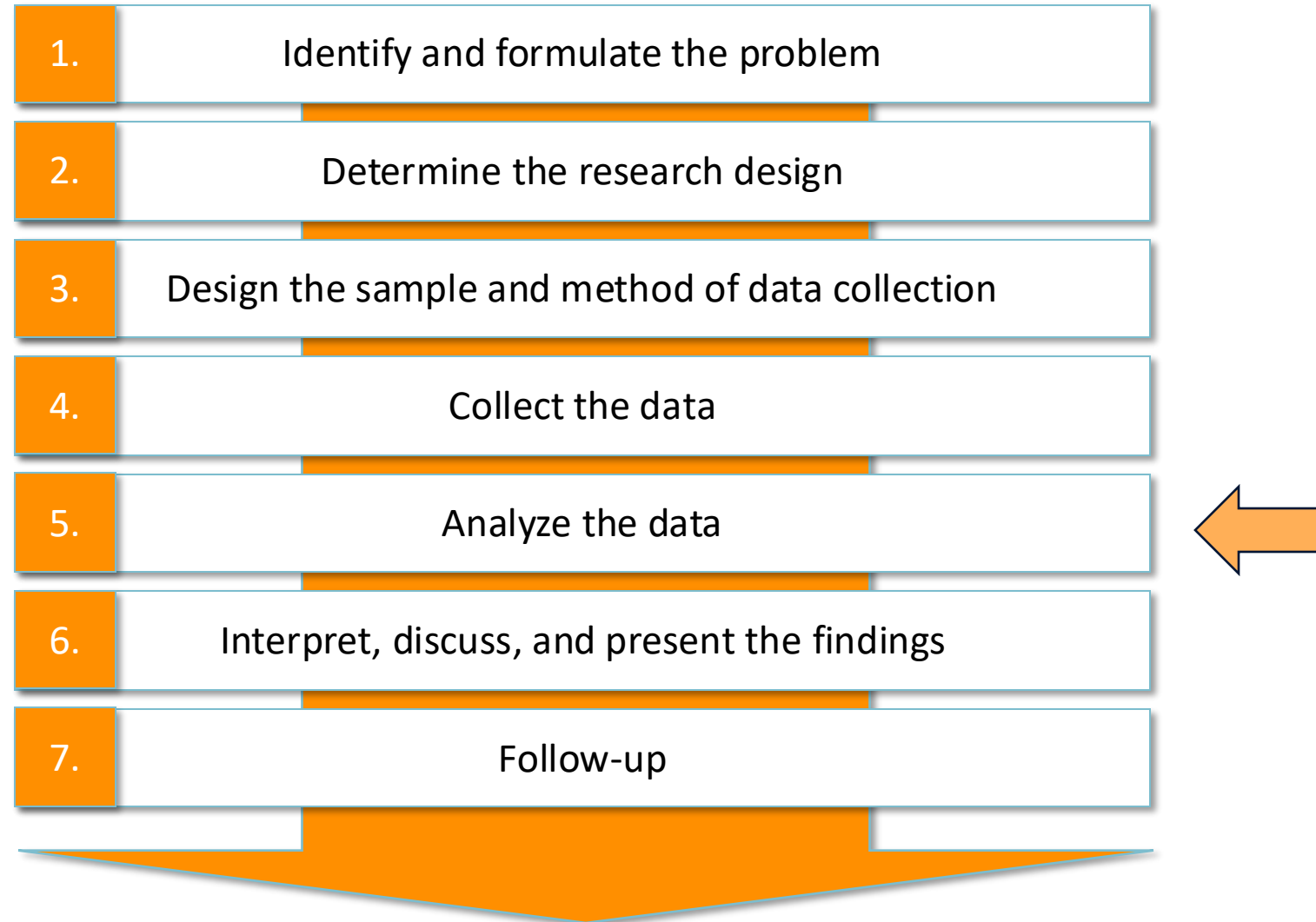
In this test we hypothesized that consumers are willing to recommend the airline to a friend or colleague. The result of a one-sample t test shows that the sample mean of willingness to recommend ($M = 8.28$) is significantly higher than the test value of 6 ($t_{1064} = 29.611, p < .001$). Hence our hypothesis was supported.

$t_{\text{degree of freedom}} = t \text{ value}$

When $p < .001$, report $p < .001$
When $.001 < p < .01$, report $p < .01$
When $.01 < p < .05$, report $p < .05$
When $.05 < p$, report $.05 < p$



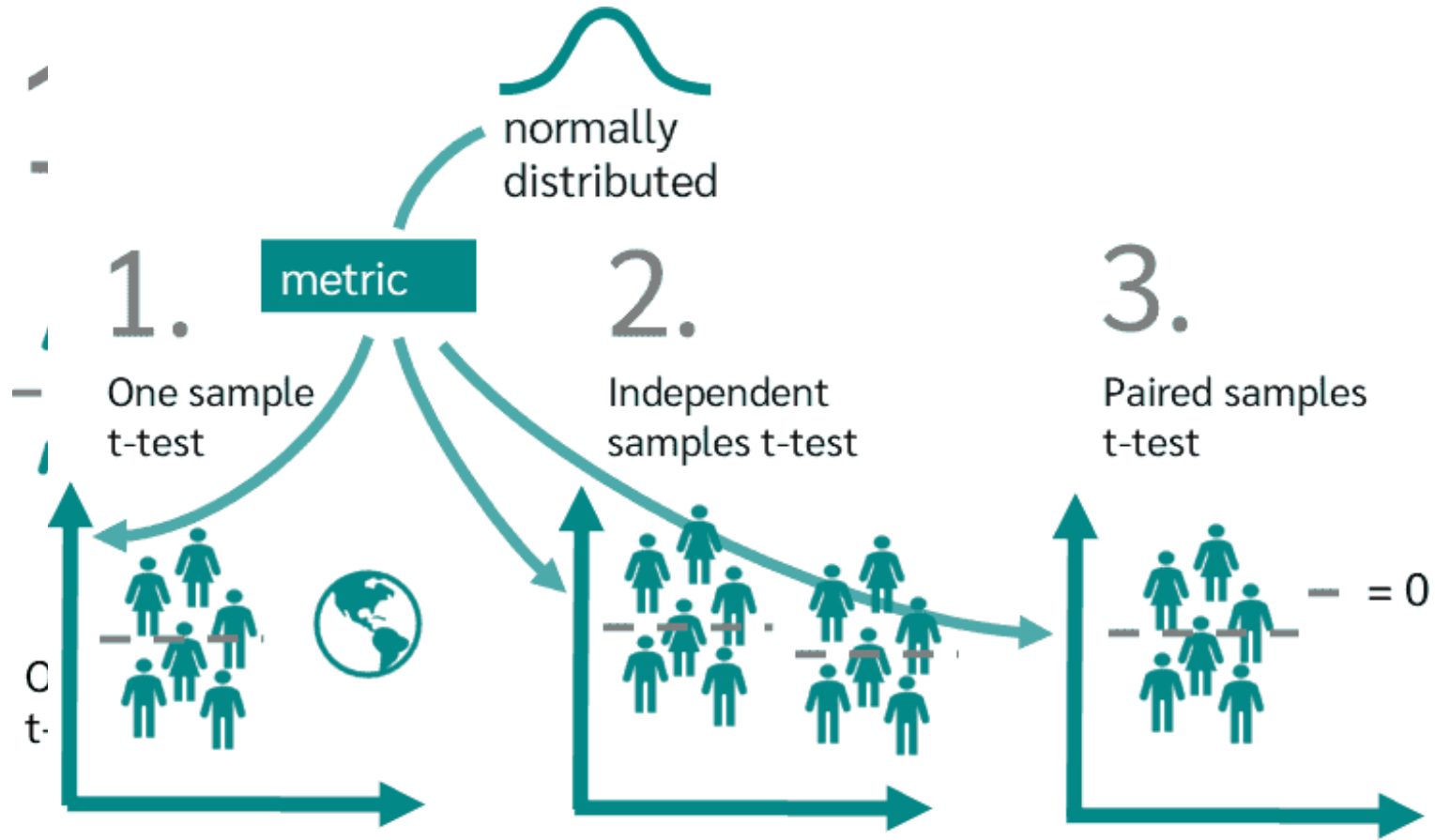
Testing for Differences



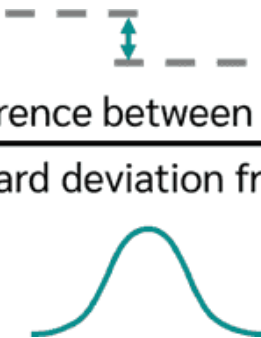
Testing for Differences

Type of Tests	Purpose of Test
One Sample T-Test	Compare the mean of a question answered by one group of subjects against a constant on metric data
Paired Samples T-Test	Compare two means of two different questions answered by the same group of subjects on metric data
Independent Samples T-Test	Compare the mean of a question answered by two different groups of subjects on metric data
ANOVA	Compare the means of a question answered by more than two different groups on metric data

Testing for Differences (t-tests)



T Statistic

$$t = \frac{\text{Difference between mean values}}{\text{Standard deviation from the mean}}$$


Standard error

One-sample t-test

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

Mean of the sample $\bar{X}$

Reference value μ

Standard deviation s

Number of cases n

Independent sample t-test

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Mean sample 1 $\bar{X}_1$

Mean sample 2 $\bar{X}_2$

Standard deviation Sample 1 and 2 s_1^2, s_2^2

Number of cases Sample 1 and 2 n_1, n_2

Paired sample t-test

$$t = \frac{\bar{X}_d - 0}{\frac{s}{\sqrt{n}}}$$

Mean of the difference $\bar{X}_d$

Standard deviation s

Number of cases n



Paired sample t-test

❑ Compare the differences in the means of *the paired samples*

Assumption

- Test variable measured on an interval or ratio scale
- Independence of observation
- $n > 30$

Table 6.4 Sales data (extended)

Store	Sales (units)	
	No point of sale display	Point of sale display
1	46	50
2	51	52
3	40	43
4	48	48
5	46	47
6	45	45
7	42	44
8	51	49
9	49	51
10	43	44

the mean difference

$$\bar{d} = \frac{1}{n} \sum_{i=1}^{10} d_i = 1.2$$

the standard error of this difference

$$s_{\bar{d}} = \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^{10} (d_i - \bar{d})^2}}{\sqrt{n}} = 0.533$$

the test statistics

$$t = \frac{\bar{d} - \mu_d}{s_{\bar{d}}} = \frac{1.2 - 0}{0.533} = 2.250$$

Critical values of the t-distribution for different significance levels α

α df	0.2	0.1	0.05	0.025	0.01	0.005
1	1.376	3.078	6.314	12.706	31.821	63.657
2	1.061	1.886	2.920	4.303	6.965	9.925
3	0.978	1.638	2.353	3.182	4.541	5.841
4	0.941	1.533	2.132	2.776	3.747	4.604
5	0.920	1.476	2.015	2.571	3.365	4.032
6	0.906	1.440	1.943	2.447	3.143	3.707
7	0.896	1.415	1.895	2.365	2.998	3.499
8	0.889	1.397	1.860	2.306	2.896	3.355
9	0.883	1.383	1.833	2.262	2.821	3.250
10	0.879	1.372	1.812	2.228	2.764	3.169
11	0.876	1.363	1.796	2.201	2.718	3.106
12	0.873	1.356	1.782	2.179	2.681	3.055
13	0.870	1.350	1.771	2.160	2.650	3.012
14	0.868	1.345	1.761	2.145	2.624	2.977
15	0.866	1.341	1.753	2.131	2.602	2.947
16	0.865	1.337	1.746	2.120	2.583	2.921
17	0.863	1.333	1.740	2.110	2.567	2.898
18	0.862	1.330	1.734	2.101	2.552	2.878
19	0.861	1.328	1.729	2.093	2.539	2.861
20	0.860	1.325	1.725	2.086	2.528	2.845
21	0.859	1.323	1.721	2.080	2.518	2.831
22	0.858	1.321	1.717	2.074	2.508	2.819
23	0.858	1.319	1.714	2.069	2.500	2.807
24	0.857	1.318	1.711	2.064	2.492	2.797
25	0.856	1.316	1.708	2.060	2.485	2.787
26	0.856	1.315	1.706	2.056	2.479	2.779
27	0.855	1.314	1.703	2.052	2.473	2.771
28	0.855	1.313	1.701	2.048	2.467	2.763
29	0.854	1.311	1.699	2.045	2.462	2.756
30	0.854	1.310	1.697	2.042	2.457	2.750

Paired sample t-test

Example

e1	Numeric	8	0	... with Oddjob Airways you will arrive on time.	{1, Very low}...
s1	Numeric	8	0	... with Oddjob Airways you will arrive on time.	{1, Very uns...

Value ▾	Label
1	Very low
100	Very high

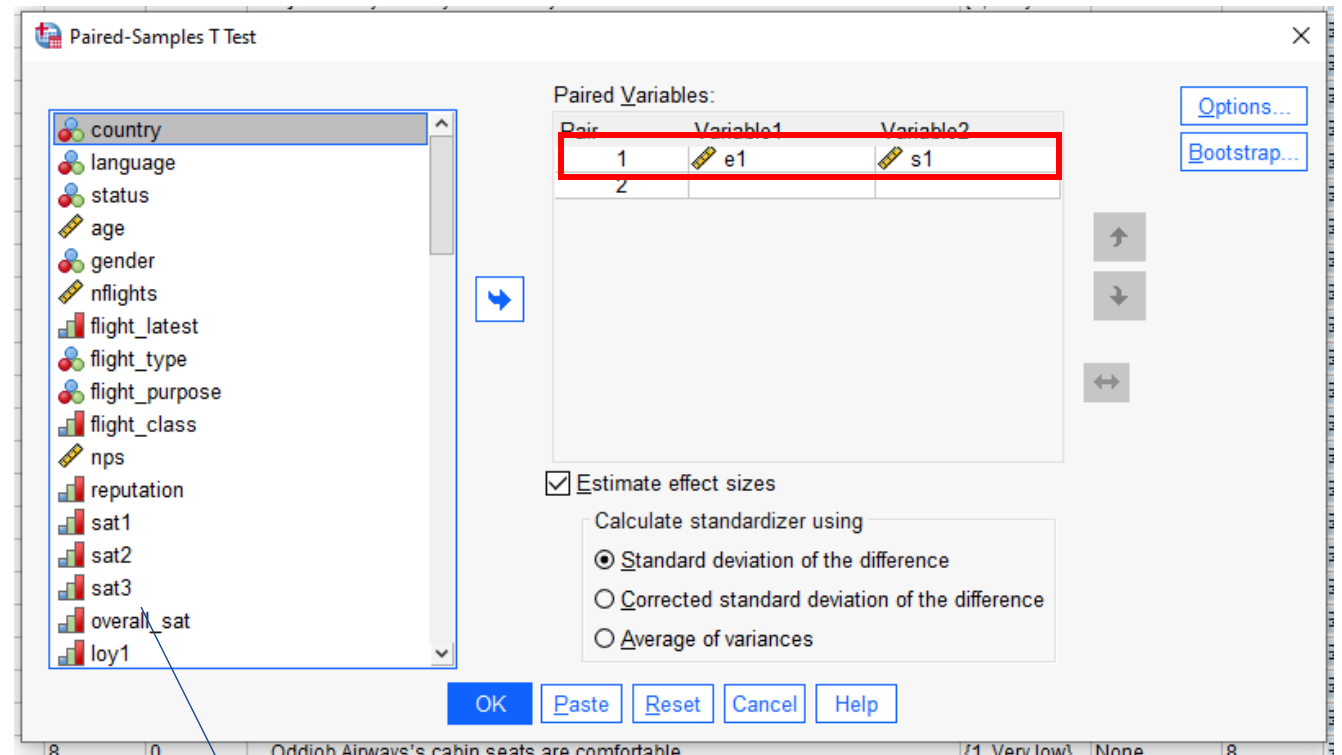
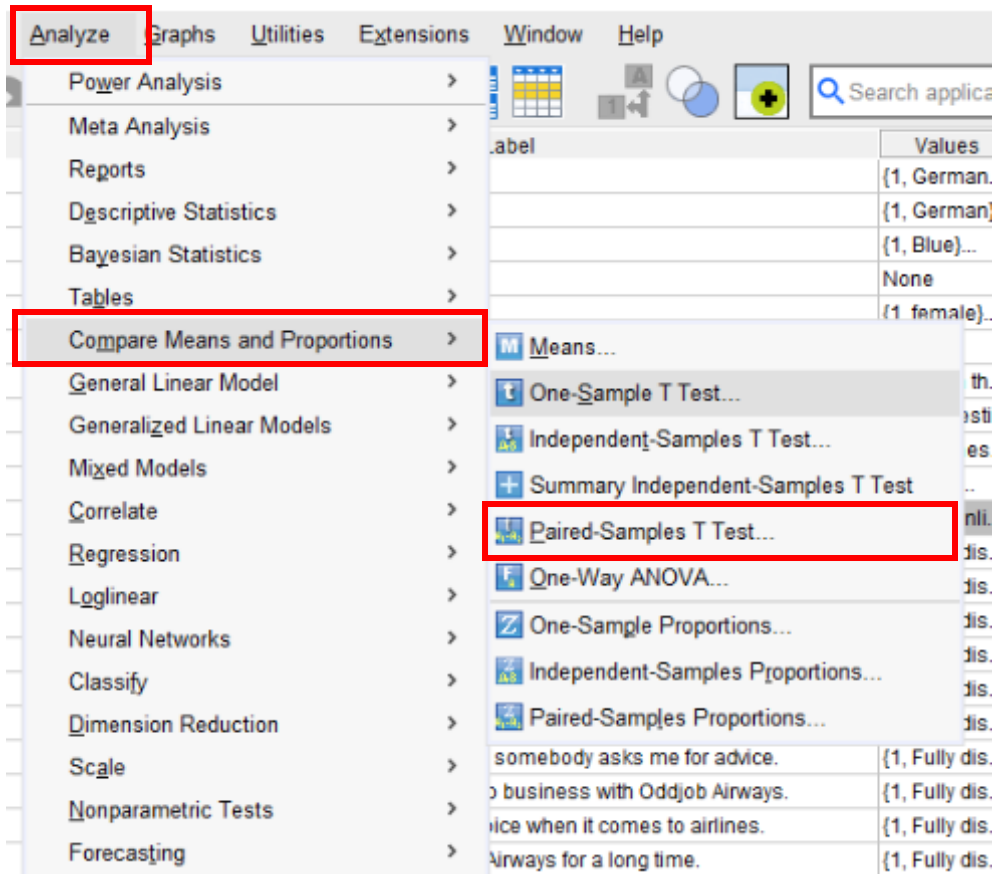
Value ▾	Label
1	Very unsatisfied
100	Very satisfied

- One group; two interval scales (both 100-points Likert scales) -> paired sample t test
- Significance level: 5%
- Testing variables: expectation of timeliness (e1), satisfaction with timeliness (s1)
- Hypothesis: Customers' expectation of timeliness at Oddjob Airways varies from their real satisfaction with the airline's timeliness.



Paired sample t-test

Analyze → Compare Means and Proportions → Paired Samples T Test → Select the variables of interest



Tip
Right click any variable and select
"display variable *names*"

Paired sample t-test

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	... with Oddjob Airways you will arrive on time. e1	86.08	1038	19.395	.602
	... with Oddjob Airways you will arrive on time. s1	60.91	1038	26.022	.808

Paired Samples Test

		Paired Differences					Significance			
					95% Confidence Interval of the Difference					
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper	t	df	One-Sided p	Two-Sided p
Pair 1	... with Oddjob Airways you will arrive on time. - ... with Oddjob Airways you will arrive on time.	25.171	27.230	.845	23.512	26.829	29.781	1037	<.001	<.001

- $\text{Mean}_{\text{Expectation}} = 86.08$, $\text{Mean}_{\text{Satisfaction}} = 60.91$
- Sig. value (p value) $< 0.001 < 0.5$
- H1 supported



Paired sample t-test

How to report the results of paired samples t-test?

Example

In this test, we hypothesized that customers' expectation of timeliness at Oddjob Airways varies from their real satisfaction with the airline's timeliness. The result of a paired samples t-test shows that customers' satisfaction with timeliness is lower than their expectation ($M_{\text{Expectation}} = 86.08$, $M_{\text{Satisfaction}} = 60.91$, $t_{1037} = 29.781$, $p < .001$). Therefore, our hypothesis was supported.

$t_{\text{degree of freedom}} = t \text{ value}$



Independent sample t-test

❑ Compare the differences in the means of the *two independent samples*

Assumption

- Is the test variable measured on at least an interval scale?
- Are the observations independent?
- n (each group) > 30

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_{\bar{x}_1 - \bar{x}_2}}$$

Examples

- Does heavy and light users' satisfaction with products differ?
- Do male customers spend more money online than female customers?
- Do US teenagers spend more time on Facebook than German teenagers?

Independent sample t-test

Example

flight_purpo...	Numeric	8	0	What was the purpose of your latest trip?	{1, Busines...
overall_sat	Numeric	8	0	Overall, I am satisfied with the price performance ratio of Oddjob Airways.	{1, Fully dis...

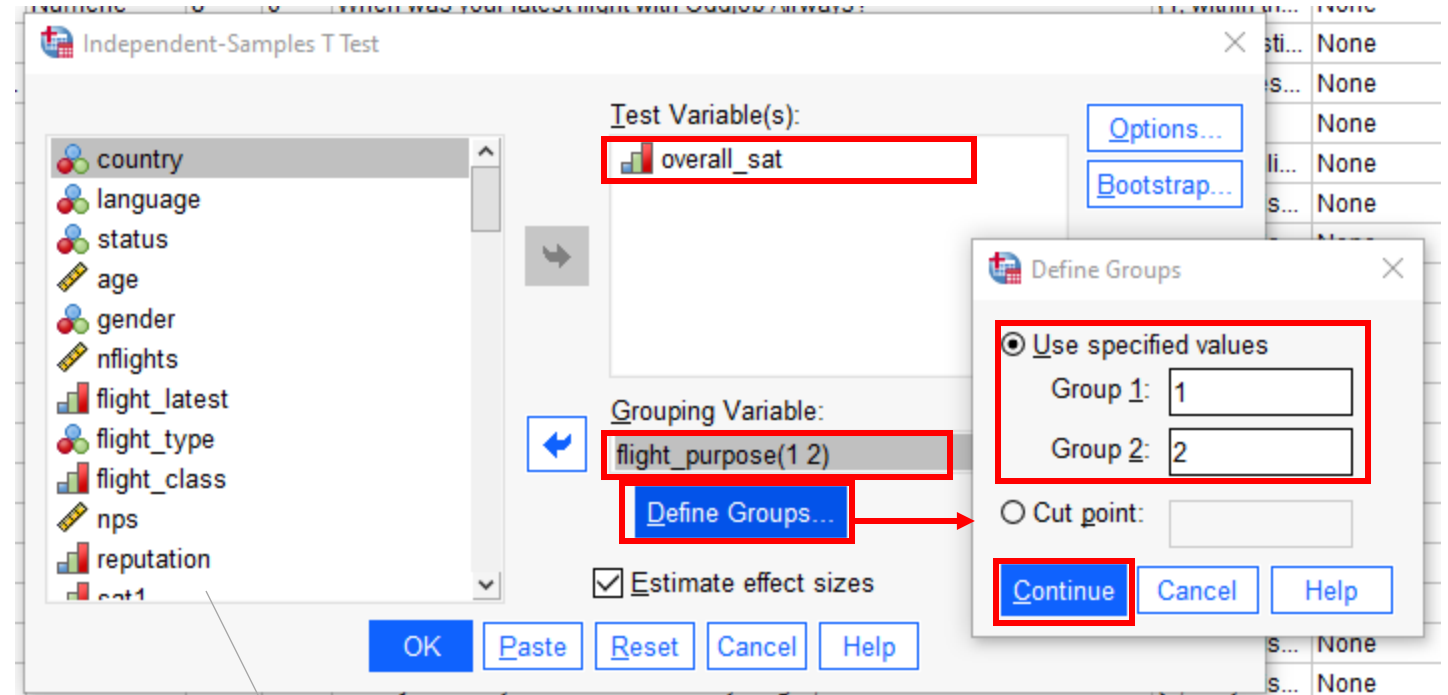
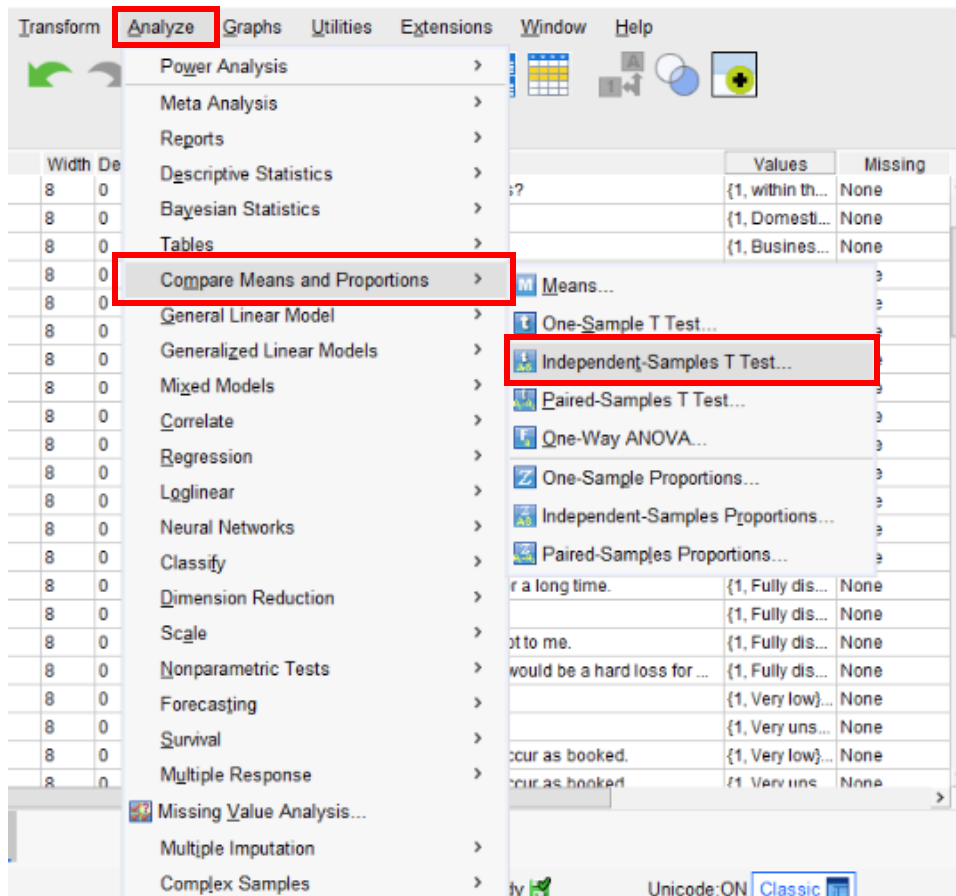
Value ▾	Label	
1	Business	+
2	Private	

Value ▾	Label	
1	Fully disagree	+
7	Fully agree	

- Two groups; one interval scale variable -> independent sample t test
- Significance level: 5%
- Grouping variable: purpose of latest trip (flight_purpose)
- Testing variable: overall satisfaction (overall_sat)
- Hypothesis: The overall satisfaction of customers with the price-performance ratio of Oddjob Airways varies between those who flew for business and those who flew for private reasons on their latest trip.

Independent sample t-test

Analyze → Compare Means and Proportions → Independent-Samples T test → Test Variable(s) = DV → Grouping variable = IV → Define Groups → Continue



Tip

Right click any variable and select "display variable names"

Independent sample t-test

Group Statistics

What was the purpose of your latest trip?		N	Mean	Std. Deviation	Std. Error Mean
Overall, I am satisfied with the price performance ratio of Oddjob Airways.	Business	525	3.99	1.541	.067
	Private	540	4.62	1.646	.071

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						95% Lower Bound
		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	
Overall, I am satisfied with the price performance ratio of Oddjob Airways.	Equal variances assumed	2.949	.086	-6.444	1063	<.001	<.001	-.630	.098	
	Equal variances not assumed			-6.450	1061.530	<.001	<.001	-.630	.098	

Check Levene's assumption test and its p value

H0: variances (Business) = variances (Private) ($p > .05$)

H1: variances (Business) $\neq$ variances (Private) ($p < .05$)



Independent sample t-test

How to report the results of independent samples t-test?

Example

In this study, we hypothesized that customers' overall satisfaction with the price-performance ratio of Oddjob Airways differs between those who traveled for business and those who traveled for personal reasons on their latest trip. The results of an independent samples t-test indicate that customers traveling for personal reasons have a higher satisfaction level with the airline's price-performance ratio compared to those traveling for business ($M_{\text{private}} = 4.62$, $M_{\text{business}} = 3.99$, $t_{1063} = -6.444$, $p < .001$). Therefore, our hypothesis is supported.

Group means

$t_{\text{degree of freedom (df)}} = t \text{ value}$